

Cultural Homophily in Tennis Doubles

Grand Slam Analysis 2018–2025

Prepared for supervisor review · May 2026 · Code: code/analysis/homophily_analysis.ipynb

Key Findings

- **Cultural homophily does not significantly predict match wins** after controlling for ranking quality and career performance. Same nationality (+3.1 pp, $p = 0.23$), same language (+3.4 pp, $p = 0.70$), and language proximity (+0.6 pp, $p = 0.95$) are all positive but statistically insignificant.
- **Same language shows a borderline pressure advantage:** teams sharing an official language are approximately 20 pp more likely to win a tiebreak ($p \approx 0.11$), consistent with communication benefits under stress — but not significant at the 5% level.
- **No Olympic build-up effect:** homophily does not predict better outcomes during the 2022–23 Paris preparation period. All interaction terms are small and insignificant.
- **Homophily shares are stable pre-2024** (~40% same-nationality, ~53% same-language from 2018–23), then drop sharply in 2024–25 — an artifact of expanded draw sizes (134 vs. ~62 matches per Grand Slam).
- **Ranking and career win rate dominate** match outcomes, as expected.

1. Data Overview

- **Matches:** 2,192 total | 2,128 Grand Slams | 64 Olympics
- **Years:** 2018, 2019, 2021, 2022, 2023, 2024, 2025 (2020 excluded — COVID)
- **Tournaments:** Australian Open, Roland Garros, Wimbledon, US Open, Olympics (2021, 2024)
- **Analysis sample:** 3,368 team-level observations (1,684 GS matches $\times$ 2 teams)

Player rankings and biographical profiles (nationality, language, coach) are sourced from ATP Tour snapshots merged to each match. Language variables derive from the CEPII Gravity dataset: *same language (official)* uses common official language; *language proximity* uses the broader ethno-linguistic criterion.

2. Pressure Outcomes

2.1 Tiebreak Observations

A **regular tiebreak** (7-point, played at 6-6 in sets 1 or 2) occurred in **44.0% of matches**. A **match tiebreak** (10-point super-tiebreak, replacing a full third set) was played in 30 matches, all at the Olympics or Wimbledon. For regression purposes both types are combined under *any tiebreak*.

Pressure Outcome	N	Share
Set-1 tiebreak (7-pt)	530	24.2%
Set-2 tiebreak (7-pt)	569	26.0%
Set-3 regular tiebreak (7-pt)	230	10.5%
Match tiebreak / super-tb (10-pt, set 3 ≥ 8)	30	1.4%
Any tiebreak (all types)	1,076	49.1%
Match went to 3 sets	903	41.2%
Comeback win (winner lost set 1)	395	18.0%

N = 2,192 total matches. Regular tiebreaks are most frequent at Wimbledon (52%) and least frequent at Roland Garros (35%).

2.2 By Tournament

Tournament	N Matches	Reg. Tiebreaks	TB Rate	Match Tiebreaks	Comeback Wins	Comeback Rate
Australian Open	580	262	45.2%	0	109	18.8%
Olympics	64	28	43.8%	20	13	20.3%
Roland Garros	581	201	34.6%	0	114	19.6%
US Open	375	165	44.0%	0	56	14.9%
Wimbledon	592	308	52.0%	10	103	17.4%

2.3 Comeback Wins

A comeback win is defined as the eventual match winner having lost the first set. This occurred in **395 matches (18.0%)** of the full sample, and in **43.7%** of three-set matches — meaning the team that won the first set subsequently lost the match roughly two-fifths of the time.

3. Cultural Homophily Across Olympic Cycles

The descriptive table below shows the share of pairs (averaged over both teams in each match) with each homophily characteristic, broken out by Olympic cycle. Grand Slam matches only — the Olympics tournament is excluded because same-nationality pairing is compulsory there.

	Same Nationality (%)	Same Language (%)	Ling. Proximity (%)	N matches
Full Sample	37.3	50.2	50.7	2128.0
Pre-Tokyo (2018–19)	40.0	53.4	54.3	512.0
Post-Tokyo (2021)	40.4	54.2	54.2	249.0
Pre-Paris (2022–23)	39.8	53.6	54.6	487.0
Paris 2024	30.9	45.7	45.2	440.0
Post-Paris (2025)	35.9	44.8	45.7	440.0

2024–25 sample sizes are larger (134 matches per Grand Slam vs. ~62 previously) because of expanded draw coverage. This compositional shift — more early-round, lower-ranked, diverse-nationality pairs — accounts for most of the observed decline in 2024–25 homophily shares.

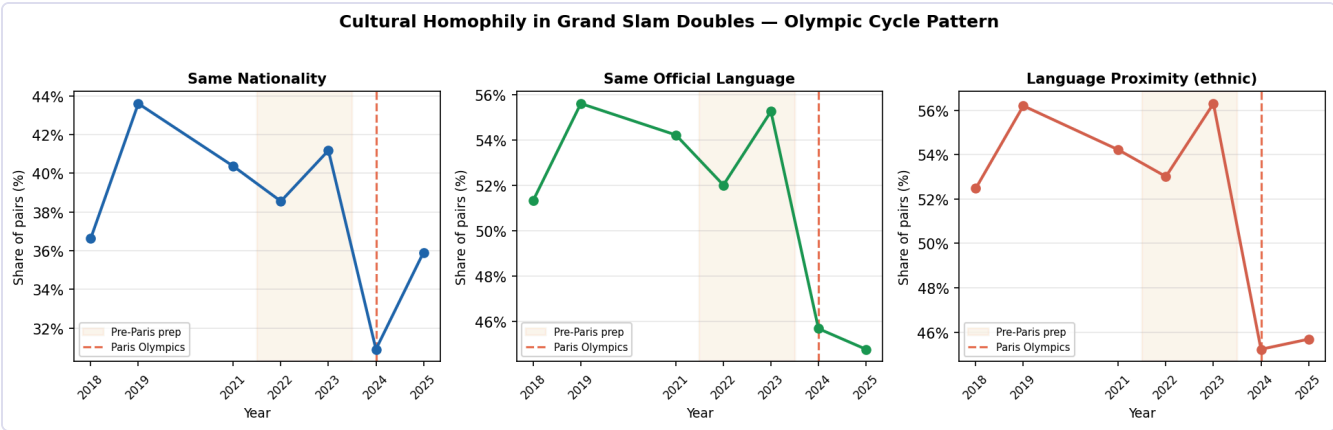


Figure 1. Share of pairs with each cultural characteristic, by year. Grand Slams only. Shaded area = pre-Paris build-up (2022–23). Red dashed line = Paris Olympics 2024.

3.1 Winner vs. Loser Team Homophily

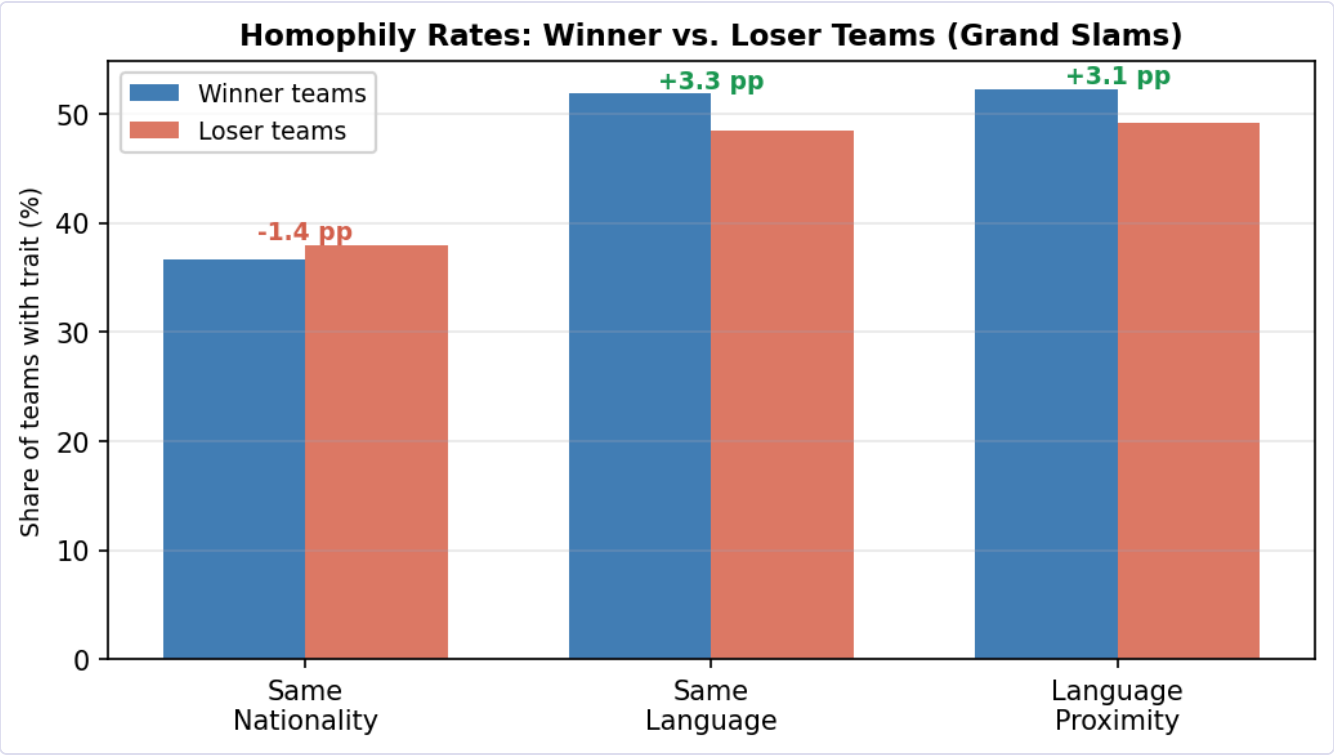


Figure 2. Percentage of teams with each homophily trait, split by match outcome. Annotation shows the winner-minus-loser gap.

Winner teams are more likely to share a language (++3.3 pp) and to be linguistically proximate (++3.1 pp) than loser teams. Same-nationality is marginally more common among losers (38.0% vs 36.6%).

4. Baseline Regressions

Specification: each match contributes two team-level observations (winner = 1, loser = 0). Regressors include the three cultural homophily dummies, log of the team's mean ATP doubles ranking, within-team career W-L ratio difference, and indicators for same dominant hand and same coach. Fixed effects for tournament, round, and year. Standard errors clustered by match.

4.1 Match Win

	LPM		Logit	
Same nationality	+0.0306	(0.0254)	+0.1365	(0.1145)
Same language (official)	+0.0341	(0.0899)	+0.1397	(0.3868)
Language proximity (ethnic)	+0.0061	(0.0886)	+0.0363	(0.3811)
Log(rank mean)	-0.1211***	(0.0080)	-0.5424***	(0.0417)
Career W-L ratio diff	-0.5581***	(0.1326)	-2.4466***	(0.6297)
Same dominant hand	+0.0177	(0.0186)	+0.0761	(0.0840)
Same coach	-0.0042	(0.0461)	+0.0032	(0.2273)
Observations	3,368		3,368	
R ² / Pseudo-R ²	0.105		0.080	

Table 3. Match Win. Outcome = 1 if team won. Tournament, round, year FE. SE clustered by match.

Significance: *** p < 0.01 ** p < 0.05 * p < 0.10 Standard errors in parentheses.

All three cultural homophily variables are positive but statistically insignificant. Ranking (log mean) and career performance are the dominant, highly significant predictors.

4.2 Tiebreak Win (pressure outcome)

Sample restricted to matches where a regular tiebreak occurred in set 1 or 2 (N = 765 matches). Outcome: team won at least one tiebreak set.

	LPM		Logit	
Same nationality	+0.0301	(0.0408)	+0.1229	(0.1693)
Same language (official)	+0.2043	(0.1277)	+0.8872	(0.6226)
Language proximity (ethnic)	-0.1847	(0.1248)	-0.8065	(0.6114)
Log(rank mean)	-0.0529***	(0.0135)	-0.2186***	(0.0574)
Career W-L ratio diff	-0.3191	(0.2488)	-1.2974	(1.0315)
Same dominant hand	+0.0494*	(0.0285)	+0.2035*	(0.1176)
Same coach	+0.0106	(0.0688)	+0.0547	(0.3009)
Observations	1,530		1,530	
R ² / Pseudo-R ²	0.028		0.020	

Table 4. Tiebreak Win. Outcome = 1 if team won a regular tiebreak. Sample: matches with tiebreak only.

Significance: *** p < 0.01 ** p < 0.05 * p < 0.10 Standard errors in parentheses.

Same language has a positive effect (+20 pp in LPM, p ≈ 0.11) and language proximity a negative one (−18 pp, p ≈ 0.14). The two measures are highly correlated, making precise separation difficult; the pattern is consistent with shared language aiding communication under tiebreak pressure, but neither reaches the 5% threshold.

4.3 Comeback Win (pressure outcome)

Sample: 657 three-set Grand Slam matches. Outcome = 1 if the eventual winner came back from losing set 1. Regressors are homophily *differences* (winner team minus loser team). HC1 heteroskedasticity-robust SE.

	LPM		Logit	
Same nationality (winner – loser)	+0.0446	(0.0410)	+0.1969	(0.1821)
Same language (winner – loser)	+0.0096	(0.1431)	+0.0530	(0.5957)
Lang. proximity (winner – loser)	-0.0429	(0.1389)	-0.2020	(0.5752)
Log(rank) advantage	+0.0218*	(0.0127)	+0.0950*	(0.0532)
Observations	657		657	
R ² / Pseudo-R ²	0.066		0.050	

Table 5. Comeback Win. Outcome = 1 if winning team came back from losing set 1. 3-set matches only. HC1 SE.

Significance: *** p < 0.01 ** p < 0.05 * p < 0.10 Standard errors in parentheses.

No homophily variable is significant for comebacks. The modest ranking advantage of the winner team is marginally predictive (p ≈ 0.09), consistent with higher-ranked teams being better able to recover from a set down.

5. Olympic Interaction

Hypothesis: players may strategically choose same-country partners during Olympic preparation windows (2022–23 for Paris 2024) because Olympic doubles requires same-country pairs. If so, homophilous teams would be better-trained for this format and should outperform in Grand Slams during that period.

Specification: each homophily variable is interacted with `pre_olympic` (= 1 for 2022–23). Year fixed effects are excluded to avoid perfect collinearity with the `pre_olympic` dummy; tournament and round FE are retained.

	LPM		Logit	
Same nationality	+0.0313	(0.0283)	+0.1399	(0.1275)
Same language	-0.0026	(0.1066)	-0.0154	(0.4429)
Language proximity	+0.0404	(0.1058)	+0.1779	(0.4389)
Pre-Olympic period (2022–23)	+0.0134	(0.0223)	+0.0541	(0.0988)
Same nationality × Pre-Olympic	-0.0073	(0.0562)	-0.0427	(0.2606)
Same language × Pre-Olympic	+0.1094	(0.1935)	+0.5163	(0.9158)
Lang. proximity × Pre-Olympic	-0.1086	(0.1907)	-0.4964	(0.9020)
Log(rank mean)	-0.1209***	(0.0081)	-0.5414***	(0.0418)
Career W-L ratio diff	-0.5478***	(0.1309)	-2.3856***	(0.6200)
Same dominant hand	+0.0180	(0.0186)	+0.0786	(0.0841)
Same coach	+0.0034	(0.0449)	+0.0395	(0.2227)
Observations		3,368		3,368
R ² / Pseudo-R ²		0.105		0.080

Table 6. Olympic Interaction. Year FE excluded (collinear with pre_olympic). Tournament and round FE retained.

Significance: *** p < 0.01 ** p < 0.05 * p < 0.10 Standard errors in parentheses.

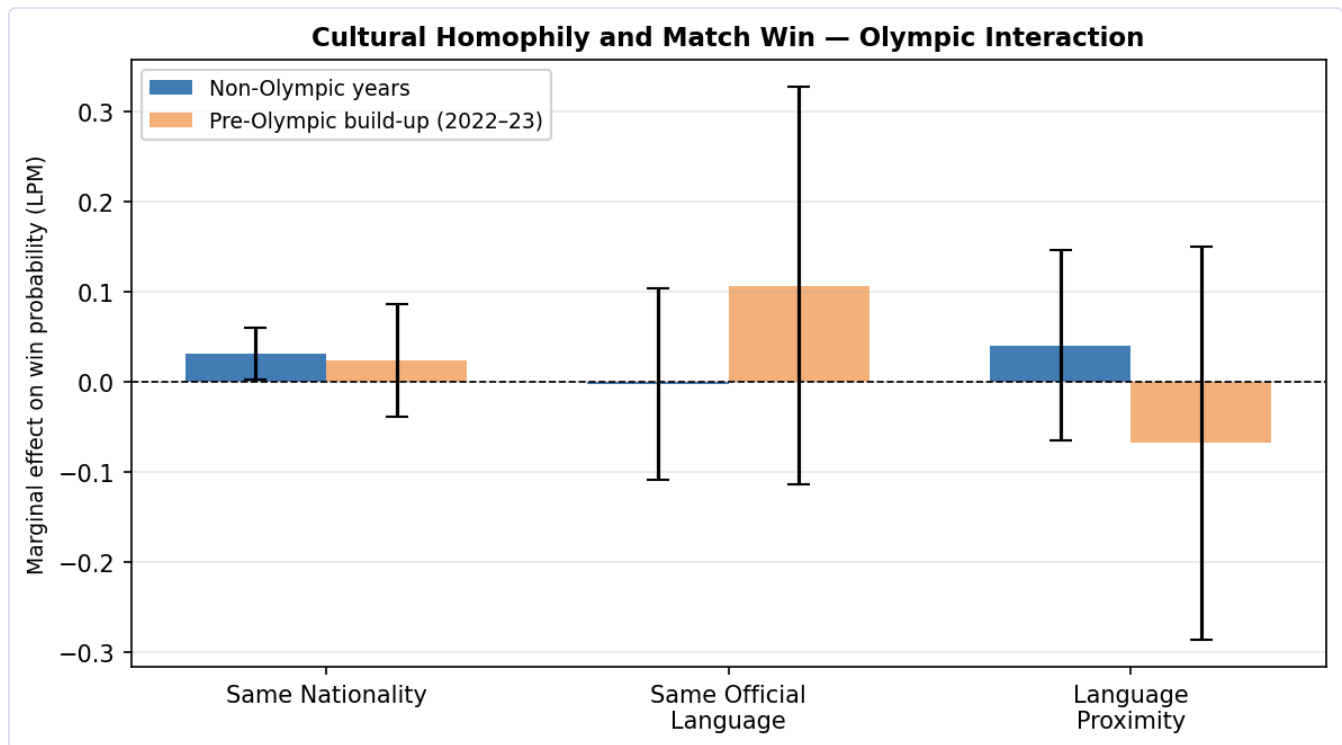


Figure 3. LPM marginal effects of each homophily variable, separately for non-Olympic years (blue) and the pre-Paris build-up period 2022–23 (orange). Error bars = ± 1 SE.

None of the interaction terms are statistically significant. The pre-Olympic dummy itself is also insignificant ($p = 0.55$), indicating no detectable change in the baseline win rate during the build-up period. The Olympic preparation hypothesis is **not supported** in these data.

6. Discussion and Next Steps

What the data say

- Language similarity has the most consistent positive sign across all three outcomes (match win, tiebreak, comeback), but never reaches conventional significance.
- Same nationality is actually slightly negative for match win in the descriptor (losers slightly more same-nationality), and the regression coefficient, while positive, is imprecisely estimated.
- Homophily shares do not rise in the run-up to the Olympics — if anything they fall in 2024–25, driven by draw expansion.

Limitations and extensions

- **Sample size:** ~1,700 Grand Slam matches. Power is limited for detecting small homophily effects ($\leq 3\text{--}5$ pp).
- **Language collinearity:** same language and language proximity are highly correlated; consider a single index.
- **Draw expansion 2024–25:** the larger sample changes composition and may mask cycle-level patterns; restrict to comparable round ranges for a cleaner trend.
- **Conditioning on partnership formation:** selection — same-country pairs may self-select into tournaments — is not yet addressed.
- **Pair fixed effects:** a within-pair panel (same partners over time) would better isolate performance from selection.

Generated automatically from `build_report.py`. All regressions run in Python (statsmodels). Data: ATP Tour Grand Slam doubles 2018–2025 (excl. 2020).